

Microprocessors & Microcontrollers

ICS 4111: Topic 4

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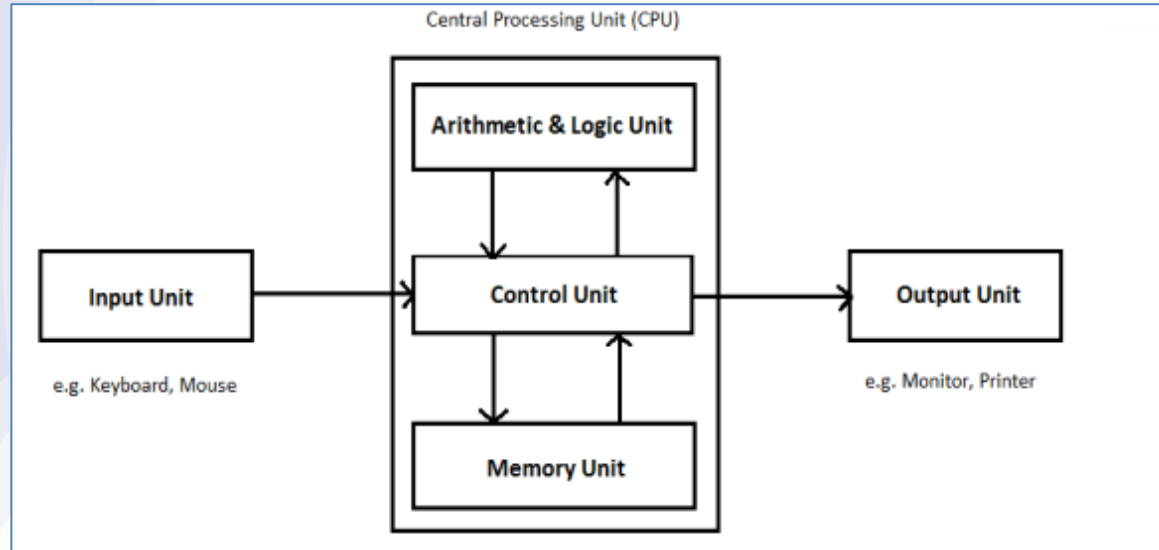
Learning Objectives

- Differentiate between microcontrollers & microprocessors.
- Discuss functionalities of microcontrollers that are in the market (Arduino, Raspberry Pi and ESP32)

Processors



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Processors are units that transform input to output.

- **ALU** – performs all logical and arithmetic operations
- **CU** – controls computer operations
- **Memory unit** – stores data & instructions

Processors

Time taken by a processor to complete a program is determined by:

- *Number of program instructions*
- *Average number of clock cycles required to execute an instruction*
- *Clock cycle time*

Processors can be categorised by:

- **Purpose:** general purpose vs application-specific
- **Size:** Small Scale Integration (SSI), Medium Scale Integration (MSI), Large Scale Integration (LSI), Very Large Scale Integration (VLSI)



Microprocessors

- **Types of Processors:** Microprocessors, Microcontroller, Digital signal processor, Media processor
- **A Microprocessor** – a VLSI programmable chip containing a CPU.
- Basic characteristics of microprocessors:
 - **Instruction set:** set of instructions it can execute
 - **Bandwidth:** number of bits that can be processed in a single instruction
 - **Clock speed:** number of instructions it can execute per second

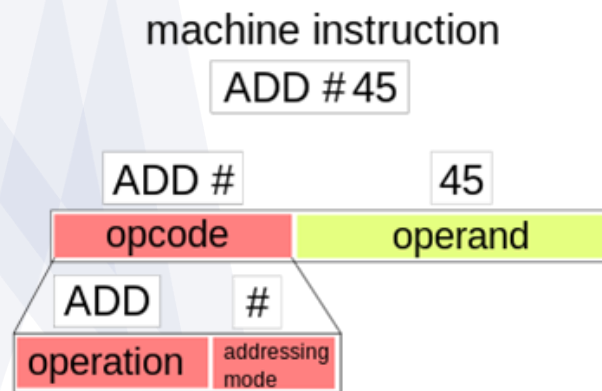


Instruction Sets

- This is a specification of opcodes in assembly language & native commands implemented by a particular processor. Opcodes operate on operands.
- A processor's instruction set specifies the basic operations supported by the machine such as:
 - **Data transfer:** moving data around memory, processor, registers, I/O e.g. MOV,
 - **Flow control:** directs execution flow of a program rather than manipulation of data e.g. JUMP, BRANCH, IF-THEN
 - **Arithmetic & Logic:** tells the control unit to request for operations within the ALU. e.g. ADD, OR, XOR, SHIFT

Instruction Sets

- This is a specification of opcodes in assembly language & native commands implemented by a particular processor.
- Opcodes operate on entities known as operands, where for instance a 32-bit operand is split into 2 fields the operation to be performed and item



(Image Source: [Computing 2 – Robert Murray, 2014](#))

Classification of Microprocessors



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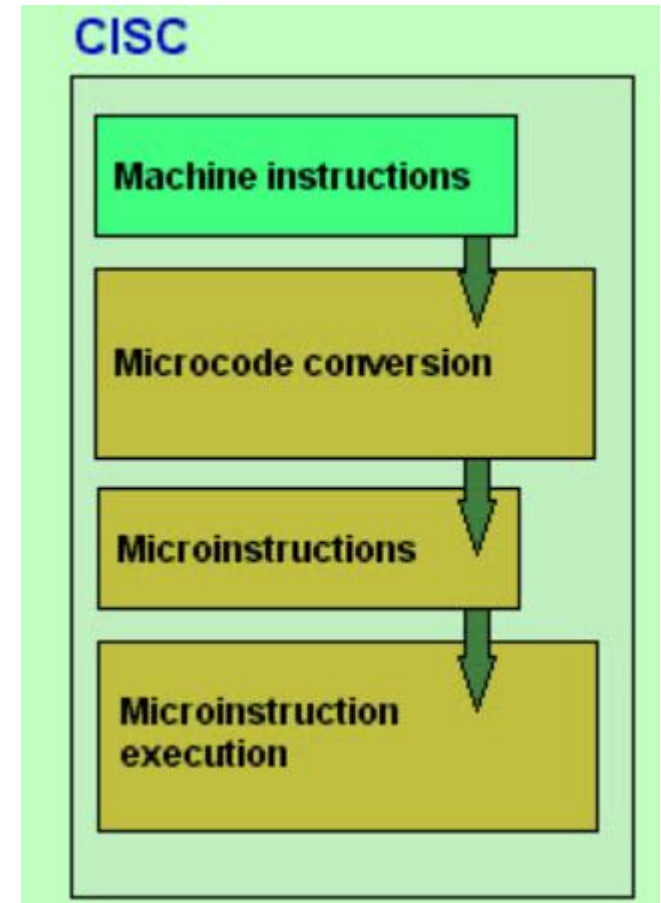
Complex Instruction Set Computer (CISC)

Complex instructions with several steps/constructs which shortens the program

Reduces number of program instructions by providing special instructions capable of performing complex operations.

Instructions completed in 2–10 machine cycles

Easy to program as it shifts burden of generating machine instructions to processor; simplifies program compiler



(Image Source:
[WatElectronics, 2021](#))

Classification of Microprocessors



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Reduced Instruction Set Computer (RISC)

Simple instructions to achieve small tasks, enables CPU to execute without using much memory

Reduces average number of clock cycles to execute an instruction

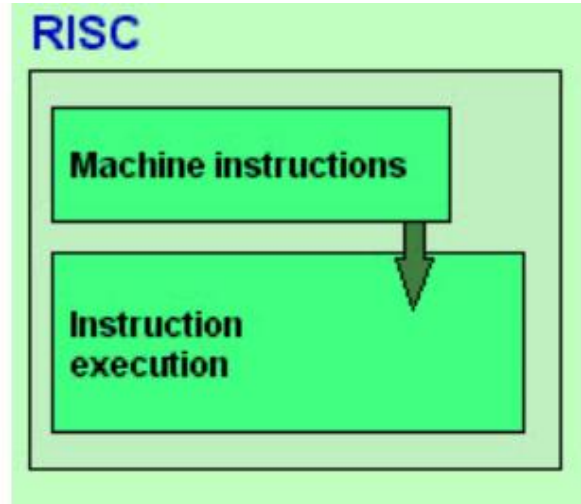
1 machine cycle to complete most instructions

Requires fewer transistors

Faster performance than CISC: few clock cycles/instructions & simpler instructions

Register allocation scheme is more complex due to large register set, increases compiler complexity

Easier to design and produce.



(Image Source:
[WatElectronics, 2021](#))

That's Part 1!

Any Questions?



Scan this code as you take a quick break



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Part 2: Microcontrollers!



Microcontrollers

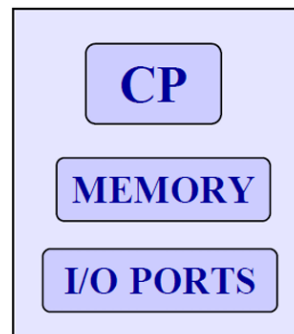
- **Controller:** a device that controls the transfer of data from a computer to a peripheral device and vice versa.
- **Microcontroller:** a VLSI unit that contains all the components comprising a controller:
 - CPU
 - RAM
 - ROM
 - I/O Ports
 - Timers
- Microcontrollers are best suited for real-time control applications



Microcontrollers vs Microprocessors

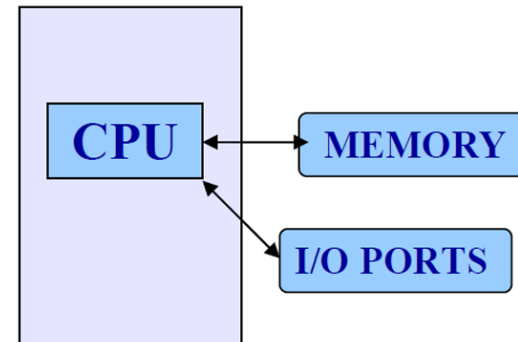
MICRO CONTROLLER

- It is a single chip
- Consists Memory, I/o ports



MICRO PROCESSOR

- It is a CPU
- Memory, I/O Ports to be connected externally



- A GP computer has similar components to a microcontroller, but it is designed to perform more tasks than a microcontroller.
- The microprocessor is a GP chip which can be used to create a computer; and it requires additional chips to handle a task e.g. memory.
- The microcontroller is designed to be more **independent and self-contained**, functioning as a tiny dedicated computer.



Arduino Microcontrollers

- The Arduino is a programmable hardware board based on either AVR or ARM (*more powerful*) processor architecture.
- The microcontroller has a special boot loader that allows users to upload programs to the microcontroller.
 - It has digital input pins for input from switches and output to Actuators (LEDS or electrical motors)
 - It also has analog pins to accept inputs from voltage-based sensors.



Arduino Microcontrollers

- Terminologies in Arduino development:
 - **I/O board** – main microcontroller controlled from the computer ([types](#) e.g. Uno, Nano, MKR, Mini)
 - **Modules** – components that overall offer additional functionality to the I/O board e.g. Bluetooth module
 - **Shield** – an add-on board/device plugged onto the I/O board to extends its capabilities e.g. cellular shield, GSM shield,
 - **Sketch** – program written for board & shields
 - **Digital**: value is HIGH or LOW
 - **Analog** – value range is between 0 and 255

That's all for now! Any Questions?

Next – Topic 5: Sensors & Actuators